

Mini-Project Proposal: 6-bit, 1 MS/s SAR ADC

Advanced Microelectronics with Cadence — JUNIA / IEMN

Group members: Member One, Member Two

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1. Objective

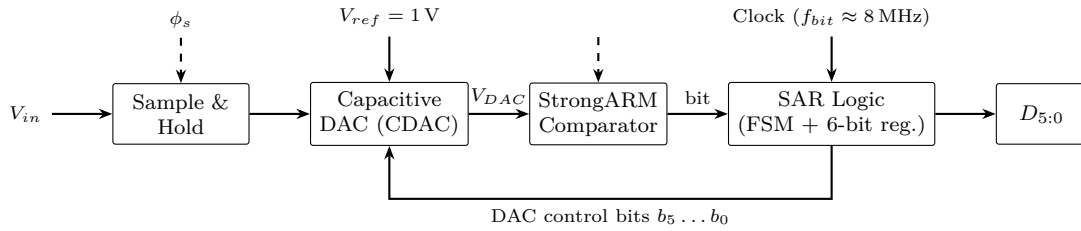
We will design and simulate a 6-bit successive-approximation-register (SAR) analog-to-digital converter. Following the course flow, we first define the specifications and architecture, then build a behavioral model and testbench, and finally replace blocks with transistor-level (analog) and RTL/gate-level (digital) implementations for mixed-signal simulation. The SAR architecture is a good fit because it combines a small analog core (comparator, capacitor array) with a digital control FSM, which exercises both halves of the course.

2. Target Specifications

Parameter	Value	Parameter	Value
Resolution	6 bits	Sampling rate	1 MS/s ($T_s = 1 \mu\text{s}$)
Reference voltage	$V_{ref} = 1 \text{ V}$	Input range	0–1 V (single-ended)
LSB size	$V_{ref}/2^6 = 15.6 \text{ mV}$	Ideal SQNR	$6.02N + 1.76 \approx 37.9 \text{ dB}$

3. Proposed Architecture

The input is sampled onto a binary-weighted capacitive DAC (CDAC). On each clock cycle a StrongARM comparator compares the CDAC output with the reference level and the SAR logic keeps or drops the bit under test, performing a binary search from MSB to LSB. With 1 sampling cycle plus 6 comparison cycles, we plan an internal bit clock of about 8 MHz (125 ns period), giving some margin for settling within the $1 \mu\text{s}$ conversion period.



Blocks:

- **CDAC:** binary-weighted array ($32C, 16C, 8C, 4C, 2C, C$ plus a dummy C , $64 C_u$ total); unit capacitor sized for matching and kT/C noise well below $\frac{1}{2}$ LSB.
- **Comparator:** StrongARM latch (dynamic, clocked, low static power), with input-referred offset below $\frac{1}{2}$ LSB.
- **Sample & Hold:** MOS switch sampling V_{in} onto the CDAC top plate (charge injection and settling to be checked).
- **SAR logic:** synchronous FSM written in HDL (SystemVerilog/Verilog), synthesized to gates.
- **Clock:** external ideal source, with non-overlapping phases for sample and convert.

4. Work Plan and Deliverables

1. **Specifications and architecture** (this proposal).
2. **Behavioral model** of the full ADC (Verilog-A / Verilog-AMS) and testbench: ramp and sine inputs.
3. **Block-level design and simulation** (at least one block, as required): StrongARM comparator and CDAC in Virtuoso (transient, offset, and speed); SAR FSM in RTL with Xcelium simulation.
4. **Mixed-signal integration:** replace behavioral blocks with the transistor-level and gate-level netlists.
5. **Performance extraction** (optional): offset, gain error, DNL, INL, SNR/ENOB from a coherent-sine or ramp test.
6. **Final presentation** (W46) and report.

Success criteria: correct 6-bit output codes across 0–1 V at 1 MS/s, with $|\text{DNL}|, |\text{INL}| < 1 \text{ LSB}$ in simulation.